TCA

ML & Data analysis report about

Titanic crisis

The RMS Titanic was a British passenger ship of approximately 46,000 tons that sank in 1912 after colliding with an iceberg during its maiden voyage. The disaster involved around 2,224 passengers and crew, of whom approximately 706 survived.

In this analysis, my peer and I aim to explore the factors that influenced survival rates on the Titanic. We investigate relationships between survival and variables such as passenger class, age, sex, fare, and embarkation point.

This study uses exploratory data analysis (EDA) and data visualization techniques to identify patterns and correlations within the dataset.

The dataset used in this analysis is the Titanic dataset, which contains information about passengers aboard the RMS Titanic.
The dataset includes approximately 891 passenger records and 12 features describing passenger characteristics such as age, sex, passenger class, fare, and embarkation point. The target variable is “Survived”, which indicates whether a passenger survived the disaster (1) or not (0).
Some features contain missing values, particularly in the Age and Cabin columns, which require handling during data cleaning. The dataset is suitable for exploratory data analysis and classification tasks.

To gain clear, factual insights we cleaned our dataset “train.csv” by the following methods:

1. Transforming column “Age” from float⟶int

```
df["Age"] = df["Age"].round(0).astype("Int64")
```

2. Dropping “SibSp” & “Parch” Columns

```
df["FamSize"] = df["SibSp"] + df["Parch"] + 1
df = df.drop(columns=["SibSp","Parch"])
```

3. Dropping “Cabin” Column where 77.1% is null values

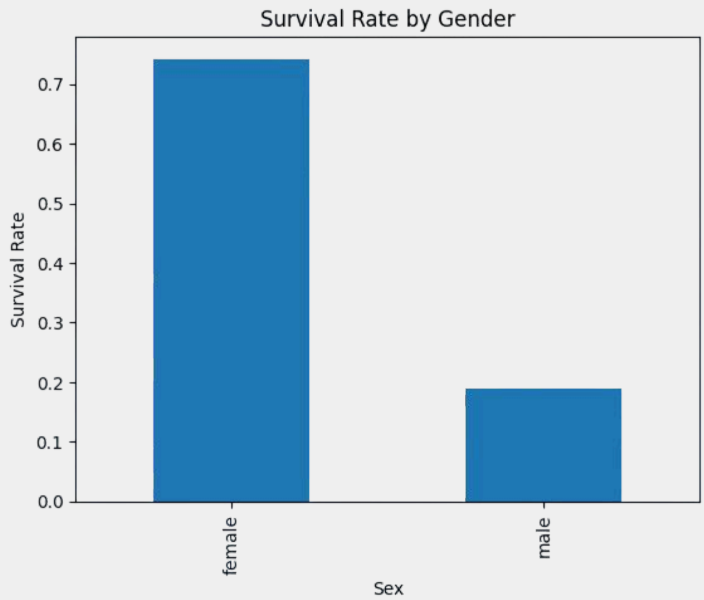
```
df = df.drop(columns="Cabin")
```

4. Handling “Age” missing values with median imputation

```
df["Age"] = df["Age"].fillna(
    df.groupby(["Sex", "Pclass"])["Age"].
   transform("Median")
)
```

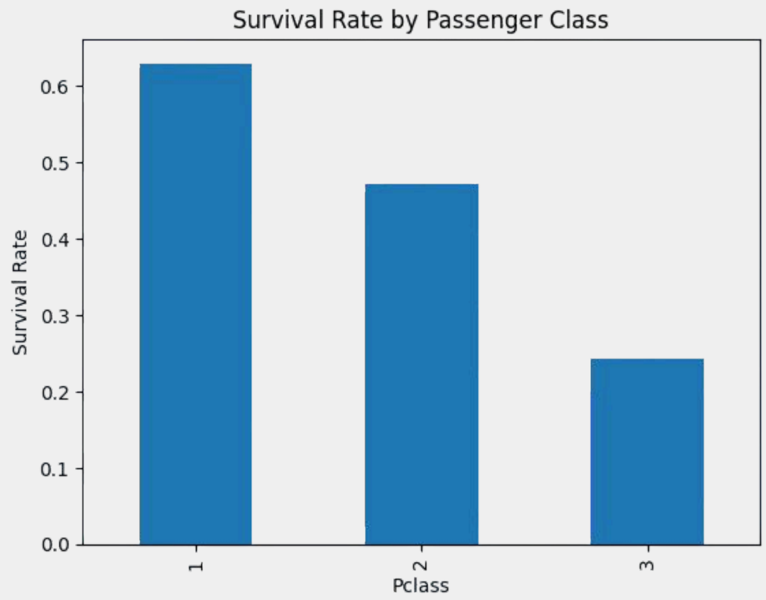
EDA was used to reveal the patterns associated with higher survival rates

1. Gender and Survival



This bar chart showed significant difference between female and male survival rate -regardless their ages- which is more than 50%

2. Passenger Class and Survival



This bar chart showed interesting difference between higher and lower classes survival rate where first class showed survival majority over second and third classes.
Moreover, as we downgrade the class, the less survival rates are.

It is conducted that:

- Gender affected survival outcomes significantly where female passengers showed higher rates of survival than male.
- Higher Pclass was associated with higher survival rates.
- Fares linked a direct relationship with survival outcomes.
- Titles extracted from names showed additional information about passengers and survival behaviour

The analysis of the Titanic dataset revealed that passenger survival was strongly influenced by a combination of demographic and socioeconomic factors. Among all features, gender and passenger class emerged as the most significant predictors of survival, with female passengers and first-class passengers having higher survival rates. Other variables such as fare and embarkation point showed relationships with survival, but these were largely influenced by their connection to passenger class rather than acting as independent factors. Feature engineering, particularly the extraction of titles from names, provided additional insight into passenger characteristics and helped improve understanding of survival patterns.